



BUILD WITH भारत 2.0

NATIONAL LEVEL HACKATHON



TEAM NAME: CODE CHAXERS

PROBLEM STATEMENT: The Accessibility Gap in Indian Public Benefits for govt. Schemes

TEAM MEMBERS: Aaditya Sharma, Ansh Sengar, Bhakti Mittal, Vidisha Kotiyal

COLLEGE: Raj Kumar Goel Institute Of Technology, Ghaziabad



PROBLEM STATEMENT

The Accessibility Gap in Indian Public Benefits for govt. Schemes

Millions of eligible Indian citizens consistently miss out on essential, life-changing public benefits due to language and structural barriers.

01 SCATTERED & FRAGMENTED PORTALS

Scheme guidelines and procedures are highly spread across many government websites (MyGov, state portals) making a single source of truth difficult to find.

02 COMPLEX LINGUISTIC BARRIERS

Official platforms, forms and scheme descriptions often use bureaucratic language that can be hard for non-English speakers and rural users. Over 70% of Indian's population speaks regional languages.

03 VULNERABLE GROUPS

Farmers, daily-wage workers, senior citizens and low-income families may struggle to navigate multiple portals and document requirements. But remain shut out due to information friction.



Core gap → Information exists, but finding the right scheme at the right time is still difficult.



SOLUTION

AI-Powered Scheme Navigator for Every Citizen

BharatSaarthi AI is a voice and text-based assistant that understands your profile — language, income, location, and needs — and instantly recommends the most relevant government schemes and benefits.

01 NATURAL LANGUAGE PROFILING

Translate or Understand conversational input language in Hindi, English, Hinglish and regional-language contexts; captures age, occupation, income and social category.

02 INSTANT ELIGIBILITY RESULTS

Predicts likely scheme qualifications with clear confidence scores and provides plain-language eligibility explanations also exact document checklist. Prioritizes schemes relevant to the user's state/region and surfaces direct links for application or further verification.

❑ Our research ensures that BharatSaarthi AI is accurate, accessible and impactful for every citizen.



"Mere papa farmer hain, unki income ₹2 lakh hai..."

User Voice Input Example

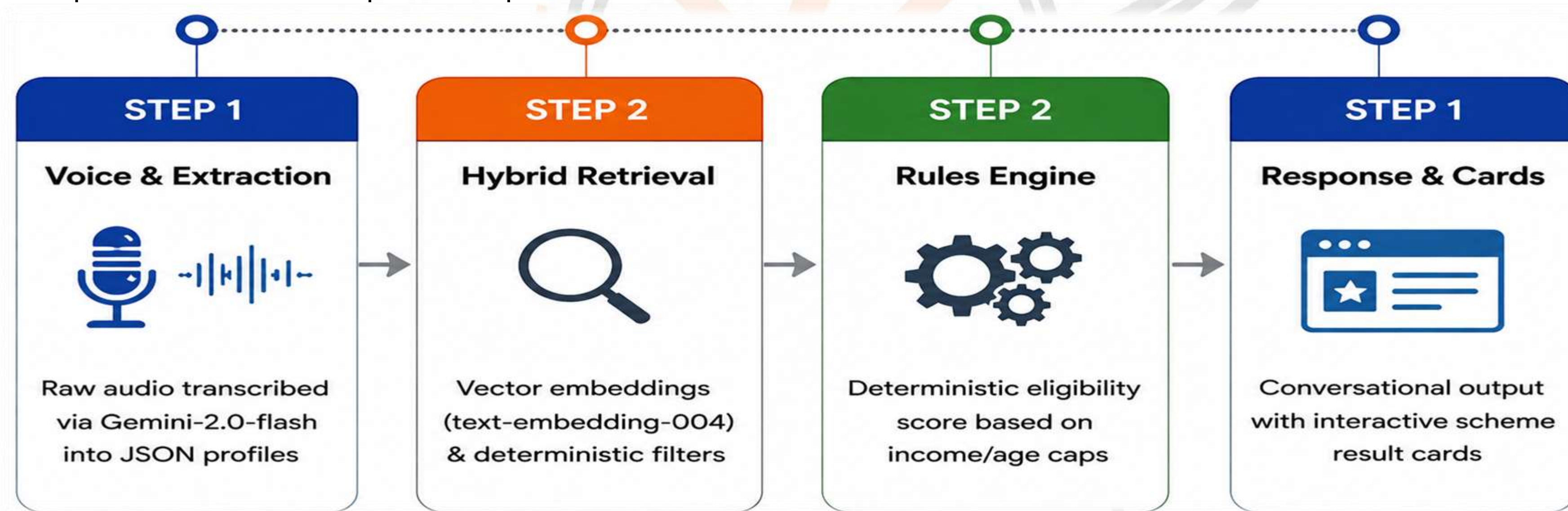




FLOW OF SOLUTION

Multi-Step Pipeline from Voice Input to Recommendation

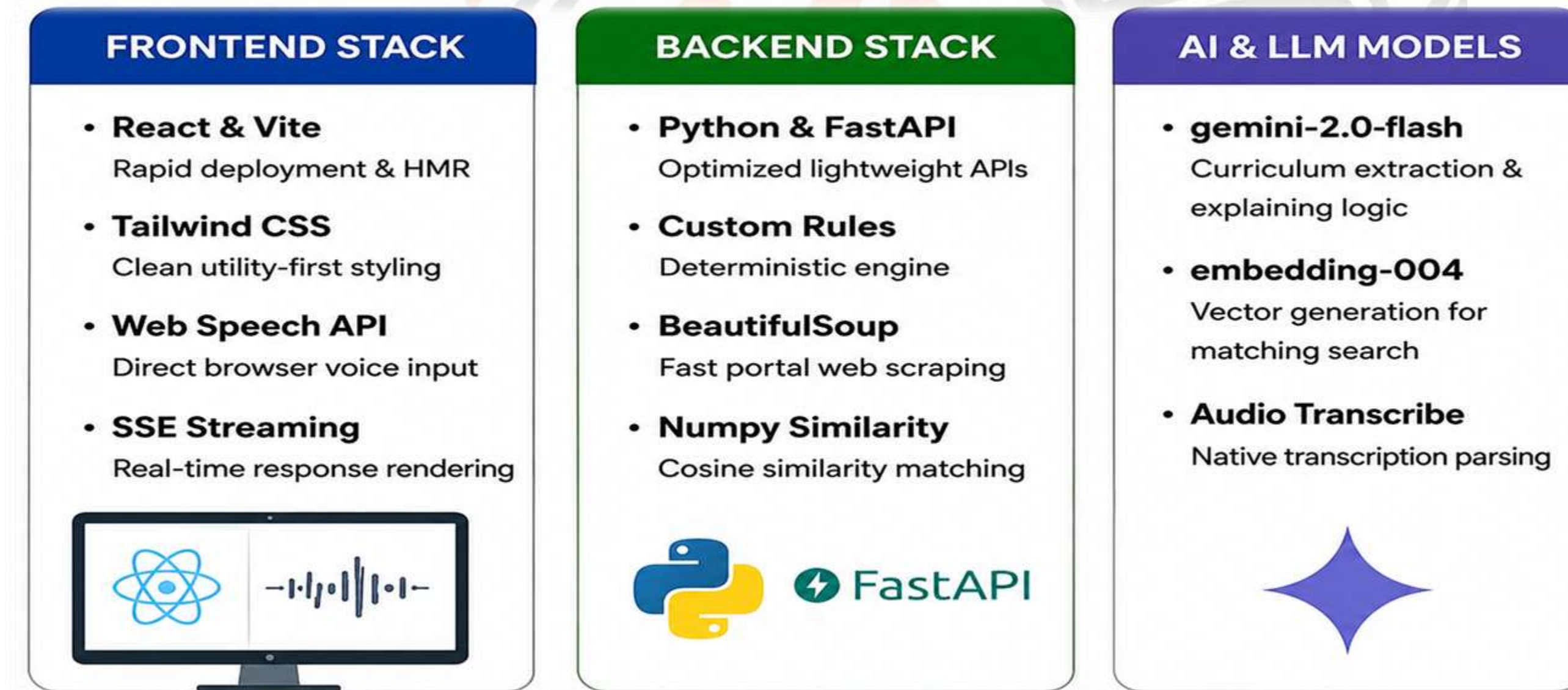
Key Highlight: Our robust, multi-step pipeline ensures, Hybrid retrieval + deterministic rules with LLM generated explanation & keep the recommendation process explainable and consistent.





TECH STACK

Technology Stack & Implementation Approach



❑ Fallback-first implementation → cached schemes remain available if live scraping or connectivity fails.



UNIQUE SELLING PROPOSITION (USP)

BharatSaarthi AI introduces unique selling propositions centered around a voice-first interface and a hybrid rules-LL engine, delivering uncompromised accuracy and unprecedented accessibility. The core innovation is **Voice-First Accessibility**, utilizing the built-in Web Speech API for seamless voice input and output. This allows the system to instantly translate Hinglish and regional speech, effectively removing basic literacy as a barrier to discovering public benefits.

The platform's **Hybrid Processing** architecture merges a robust deterministic rules engine—which ensures absolute factual matching against government criteria—with the Gemini model, which expertly translates complex legalities and bureaucratic jargon into friendly, plain language.

Additionally, the system features **State-Aware Hybrid Retrieval**, automatically filtering public schemes based on the user's specific regional residency to ensure all recommendations are geographically relevant. Finally, the platform emphasises **Direct Actionability**. Each recommended scheme card presented to the user contains a direct link to the official source or application portal, alongside a highly simplified, easy-to-read document checklist to streamline the application process.



Voice-First Accessibility

Web Speech API
input/output for
low-literacy users



Deterministic Hybrid Match

Zero hallucination
rules engine + natural
language reasons



State-Aware Local Cache

Direct source links
and regional
geofilters



FEASIBILITY & COMPETITORS

Proactive Mitigation & Market Differentiation

COMPETITOR LANDSCAPE

- **Live scraping is brittle**
scrape once → curated JSON + cached schemes fallback
- **Gemini free-tier limits**
Flash-lite-tier-model → controlled requests + cached embeddings
- **Demo Wi-Fi unreliable**
Deploy to backend + preloaded data, Vercel (frontend), not localhost
- Scheme rules can change → versioned dataset + source links.

Platform	Language Support	AI + Rules Hybrid	Voice-First	Direct Links & Checklist
BharatSaarthi AI	✓	✓	✓	✓
India.gov.in	✗	✗	—	✗
UMANG App	✓	✗	✓	✓
State Portals	✗	✗	✗	✗

Feasibility verdict: prototype-ready with a strong fallback strategy; scale requires verified scheme data and regular rule updates



RESEARCH & REFERENCE

RESEARCH



01. USER & PROBLEM RESEARCH

- Studied barriers faced by citizens in accessing public benefits.
- Primary surveys & interviews with users from rural & semi-urban areas.
- Identified pain points in language, awareness, documentation & navigation.



02. EXISTING SYSTEM & COMPETITOR STUDY

- Analyzed government portals (MyGov, state portals) and their usability.
- Studied similar platforms and assistant tools.
- Identified gaps in accuracy, personalization and accessibility.



03. AI / MULTILINGUAL VOICE RESEARCH

- Research on multilingual NLP models & translation techniques.
- Explored voice recognition (Web Speech API) for Indian languages.
- Studied hybrid approach: Rules Engine + LLM for explainable results.



04. GOVERNMENT DATA & SCHEME RESEARCH

- Collected & analyzed information from official scheme guidelines.
- Studied eligibility rules, beneficiary categories & application processes.
- Structured data for state-wise scheme mapping.

REFERENCES



GOVERNMENT PORTALS

- <https://www.india.gov.in>
- <https://www.mygov.in>
- State Government Portals



GOVERNMENT DATA SOURCES

- data.gov.in
- Open Government Data (OGD) Platform India



RESEARCH PAPERS

- IEEE Xplore Digital Library
- SpringerLink
- ResearchGate



VOICE & NLP RESEARCH

- Google Research
- Mozilla Common Voice
- Papers on Multilingual Speech Recognition



DOCUMENTATION & TOOLS

- Gemini API Documentation
- Web Speech API (MDN)
- LangChain / Embeddings Documentation